

This is the brain of your app. It calculates the ideal ranges based on the *Instar* stage and dictates the color-coded UI.

Kotlin

```
import androidx.compose.ui.graphics.Color
import androidx.lifecycle.ViewModel
import kotlinx.coroutines.flow.MutableStateFlow
import kotlinx.coroutines.flow.StateFlow

data class AdviceState(
    val statusColor: Color = Color.Gray,
    val message: String = "Enter reading to get advice."
)

class SericultureViewModel : ViewModel() {

    private val _adviceState = MutableStateFlow(AdviceState())
    val adviceState: StateFlow<AdviceState> = _adviceState

    // Logic Engine: 5 Stages of Silkworm Growth
    fun analyzeClimate(instar: Int, temp: Float, humidity: Float) {
        // Validation Rule
        if (temp !in 10f..45f || humidity !in 0f..100f) {
            _adviceState.value = AdviceState(Color.Red, "Invalid Entry! Temp
must be between 10-45°C.")
            return
        }

        // Define ideal ranges based on Instar
        val (idealTemp, idealHum) = when (instar) {
            1, 2 -> 27.5f to 85f // Chawki rearing requires high temp/humidity
            3 -> 26.0f to 80f
            4 -> 25.0f to 75f
            5 -> 24.0f to 70f // Late age requires lower temp/humidity
            else -> 25.0f to 70f
        }

        generateAdvice(temp, humidity, idealTemp, idealHum)
    }

    private fun generateAdvice(temp: Float, hum: Float, idealTemp: Float,
idealHum: Float) {
        val tempDiff = temp - idealTemp
        val humDiff = hum - idealHum
    }
}
```

```

val statusColor: Color
var message = ""

// Temperature Logic
when {
    tempDiff > 2f -> {
        statusColor = Color.Red
        message += "CRITICAL: Temp is too high! Open windows, spread
wet gunny bags. "
    }
    tempDiff < -2f -> {
        statusColor = Color.Red
        message += "CRITICAL: Temp is too low! Use charcoal
heaters/room heaters. "
    }
    tempDiff in -2f..2f && tempDiff != 0f -> {
        statusColor = Color(0xFFFFA500) // Orange
        message += "CAUTION: Temp is drifting. Monitor closely. "
    }
    else -> {
        statusColor = Color.Green
        message += "Temp is perfect. "
    }
}

// Humidity Logic
when {
    humDiff > 10f -> message += "Humidity high! Use exhaust fans or dust
lime powder."
    humDiff < -10f -> message += "Humidity low! Spray water on the floor
or hang wet cloths."
}

if (statusColor == Color.Green && humDiff in -10f..10f) {
    message = "Optimal Environment. Great job!"
}

_adviceState.value = AdviceState(statusColor, message)
}
}

```